

Julian Asilis

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U.S. Citizen

Education

- 08/2022 – 12/2026 **Ph.D. in Computer Science**, UNIVERSITY OF SOUTHERN CALIFORNIA, Los Angeles
(Expected) Supported by [NSF Graduate Research Fellowship](#), 2024–present
Research: Statistical learning theory, deep generative models.
Advisor: Vatsal Sharan
- 2016 – 2020 **A.B. in Mathematics with High Honors**, HARVARD UNIVERSITY, Cambridge, MA
Senior Thesis: [Probability Monads](#), under Michael Hopkins.

Experience

- Summer 2025 **Machine Learning Engineer Intern**, PINTEREST, Seattle, WA
- Accelerated training of large-scale ads ranking model by 32% while improving AUC.
 - Implemented data subsampling/reweighting methods, including local case-control (LCC) and reweighted SGD, to improve efficiency and reduce training costs.
- 2021 – 2022 **Research Associate**, BOSTON COLLEGE, Chestnut Hill, MA
- Researched computable learning theory and topological measures of complexity for neural nets, leading to a publication at LICS and extended abstract at CCA.
 - Served as TA for 2 computer science courses, including writing 140 pages of notes, overseeing 7 TA's, and writing scripts for automated exam grading.
- 2020 – 2021 **Quantitative Researcher**, AQR CAPITAL MANAGEMENT, Greenwich, CT
- Refined and expanded several factors used to trade dozens of assets in fixed income.
 - Performed statistical inference and time series modeling on large panel data.
 - Wrote production code in Python and SQL.

Research

Theory papers typically use alphabetical author ordering.

- Resa: Transparent Reasoning Models via SAEs**
Shangshang Wang, [Julian Asilis](#), Ömer Faruk Akgül, Enes Burak Bilgin, Ollie Liu, Deqing Fu, Willie Neiswanger.
In *Conference on Language Modeling (COLM)*, 2026.
- Textual Steering Vectors Can Improve Visual Understanding in Multimodal Large Language Models**
Woody Gan, Deqing Fu, [Julian Asilis](#), Ollie Liu, Dani Yogatama, Vatsal Sharan, Robin Jia, Willie Neiswanger.
In *Association for Computational Linguistics (ACL)*, 2026.
- Tina: Tiny Reasoning Models via LoRA**
Shangshang Wang, [Julian Asilis](#), Ömer Faruk Akgül, Enes Burak Bilgin, Ollie Liu, Willie Neiswanger.
In *International Conference on Learning Representations (ICLR)*, 2026.

9. **Semi-Random Graphs, Robust Asymmetry, and Reconstruction**
Julian Asilis, Xi Chen, Dutch Hansen, Shang-Hua Teng.
 In *Innovations in Theoretical Computer Science (ITCS)*, 2026.
8. **On Agnostic PAC Learning in the Small Error Regime**
Julian Asilis, Mikael Møller Høgsgaard, Grigoris Velegkas.
 In *Neural Information Processing Systems (NeurIPS)*, 2025. [Spotlight paper](#)
7. **Local Regularizers Are Not Transductive Learners**
 Sky Jafar, Julian Asilis, Shaddin Dughmi.
 In *Conference on Learning Theory (COLT)*, 2025.
6. **Understanding Aggregations of Proper Learners in Multiclass Classification**
Julian Asilis, Mikael Møller Høgsgaard, Grigoris Velegkas.
 In *Conference on Algorithmic Learning Theory (ALT)*, 2025.
5. **Proper Learnability and the Role of Unlabeled Data**
Julian Asilis, Siddhartha Devic, Shaddin Dughmi, Vatsal Sharan, and Shang-Hua Teng.
 In *Conference on Algorithmic Learning Theory (ALT)*, 2025.
4. **Transductive Sample Complexities Are Compact**
Julian Asilis, Siddhartha Devic, Shaddin Dughmi, Vatsal Sharan, and Shang-Hua Teng.
 In *Neural Information Processing Systems (NeurIPS)*, 2024.
3. **Open Problem: Can Local Regularization Learn All Multiclass Problems?**
Julian Asilis, Siddhartha Devic, Shaddin Dughmi, Vatsal Sharan, and Shang-Hua Teng.
 In *Conference on Learning Theory (COLT)*, 2024. (Open problems track)
2. **Regularization and Optimal Multiclass Learning**
Julian Asilis, Siddhartha Devic, Shaddin Dughmi, Vatsal Sharan, and Shang-Hua Teng.
 In *Conference on Learning Theory (COLT)*, 2024.
1. **Computable PAC Learning of Continuous Features**
 Nathanael Ackerman, Julian Asilis, Jieqi Di, Cameron Freer, and Jean-Baptiste Tristan.
 In *Logic in Computer Science (LICS)*, 2022.

Community

- Summer 2023 **SHINE Mentor**, USC Summer High School Intensive in Next Generation Engineering (SHINE), Los Angeles, CA
- 2019 – 2020 **Math Mentor**, Harvard Gender Inclusivity in Mathematics (GIIM), Cambridge, MA
- 2018 – 2019 **Teaching Assistant**, Cambridge Math Circle, Cambridge, MA

Skills

Programming: Python

Languages: English, Spanish